

Cracking AirBench

AlphaGrad

Week 9 Update

What we are optimizing

- LLMs are good at **generating** solutions, but bad at **verifying** them
 - Many approaches have tried to answer this (ex. LLM as a Judge)
- One solution: evaluate LLMs against a **strictly verifiable** problem

Many such examples:

- Kernel speed
- Matrix multiplication algorithm speed
- Erdos problems

CIFAR-10 [AirBench](#)

CIFAR-10 Airbench

This repo contains the two fastest known algorithms ([edit: not anymore](#)) for training a neural network to 94% or 96% accuracy on CIFAR-10 using a single NVIDIA A100 GPU.

Script	Mean accuracy	Time	PFLOPs
airbench94_muon.py	94.01%	2.59s	0.29
airbench96_faster.py	96.00%	27.3s	3.1

(Timings done using `torch==2.4.1` on a 400W NVIDIA A100)

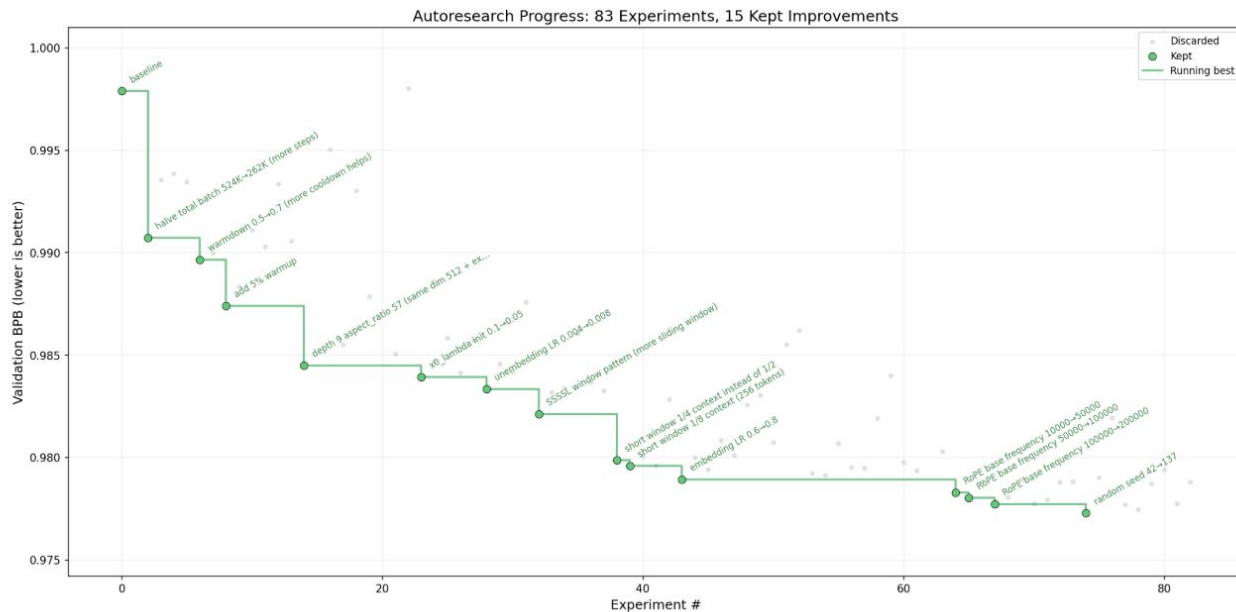
For comparison, the standard training used in most studies on CIFAR-10 is much slower:

Baseline	Mean accuracy	Time	PFLOPs
Standard ResNet-18 training	96.0%	7min	32.3

Idea: An automated AI researcher in a loop

- Karpathy's **autoresearch** repo applies this to optimizing training for [nanochat](#)
 - [Github link](#)

autoresearch



How it Works:

- Same core loop as AlphaEvolve + autoresearch: propose → evaluate (strict verifier) → keep best → iterate
- Continuously append success/failure data to `memory.md` so agent can grow from past explorations
- Use free Gemini model (`gemini-3.1-flash-lite-preview`) + A100-40GB GPU on Modal for serverless training in seconds
- Start from the best incumbent script in AirBench repo, see if we can beat the record

Initial results



Initial results

- Lowered SOTA training time to ~2.50 from ~2.59 (~2% improvement)
- But have not rigorously evaluated the solution under the same conditions (one warmup round then take average of 200x runs)

Code edits:

- enable `torch.set_float32_matmul_precision('high')`
- increase batch size 2000 -> 2500

Steps forward:

- Use multi-agent optimization loop
 - Outer coordinator suggests high level [strategy.md](#)
 - Inner agents each run suggestions (in parallel) then coordinator picks best one
- Need to better draw fine line between making not enough substantive edits (only tweaking one hyperparameter multiple times) and making too many edits (code does not compile)
- Use stronger models (GPT-5.4 level)